

Our firm entry game from last lecture

	Firm 1 enters	Firm 1 does not enter
Firm 2 enters	F1 profit = -1 F2 profit = -1	F1 profit = 0 F2 profit = 1
Firm 2 does not enter	F1 profit = 1 F2 profit = 0	F1 profit = 0 F2 profit = 0

- I claim that there is a game in mixed strategies – where each firm chooses randomly to enter or not
- Let's find those probabilities!

Our firm entry game from last lecture

	Firm 1 enters Probability that firm 1 enters = p_1	Firm 1 does not enter Probability that firm 1 does not enter = $(1 - p_1)$
Firm 2 enters Probability that firm 2 enters = p_2	F1 profit = -1 F2 profit = -1	F1 profit = 0 F2 profit = 1
Firm 2 does not enter Probability that firm 2 does not enter = $(1 - p_2)$	F1 profit = 1 F2 profit = 0	F1 profit = 0 F2 profit = 0

Our firm entry game from last lecture

	Firm 1 enters	Firm 1 does not enter
	Probability that firm 1 enters = p_1	Probability that firm 1 does not enter = $(1 - p_1)$
Firm 2 enters Probability that firm 2 enters = p_2	F1 profit = -1 F2 profit = -1	F1 profit = 0 F2 profit = 1
Firm 2 does not enter Probability that firm 2 does not enter = $(1 - p_2)$	F1 profit = 1 F2 profit = 0	F1 profit = 0 F2 profit = 0

- For this to be a Nash equilibrium, we need our players to be indifferent in expectation so:
- For Firm 1: **Expected value of red box** = **Expected value of purple box**

Our firm entry game from last lecture

	Firm 1 enters	Firm 1 does not enter
	Probability that firm 1 enters = p_1	Probability that firm 1 does not enter = $(1 - p_1)$
Firm 2 enters Probability that firm 2 enters = p_2	F1 profit = -1 F2 profit = -1	F1 profit = 0 F2 profit = 1
Firm 2 does not enter Probability that firm 2 does not enter = $(1 - p_2)$	F1 profit = 1 F2 profit = 0	F1 profit = 0 F2 profit = 0

- For this to be a Nash equilibrium, we need our players to be indifferent in expectation so:
- For Firm 1: **Expected value of red box** = **Expected value of purple box**
- For Firm 2: **Expected value of blue box** = **Expected value of green box**

Why is this a Nash equilibrium?

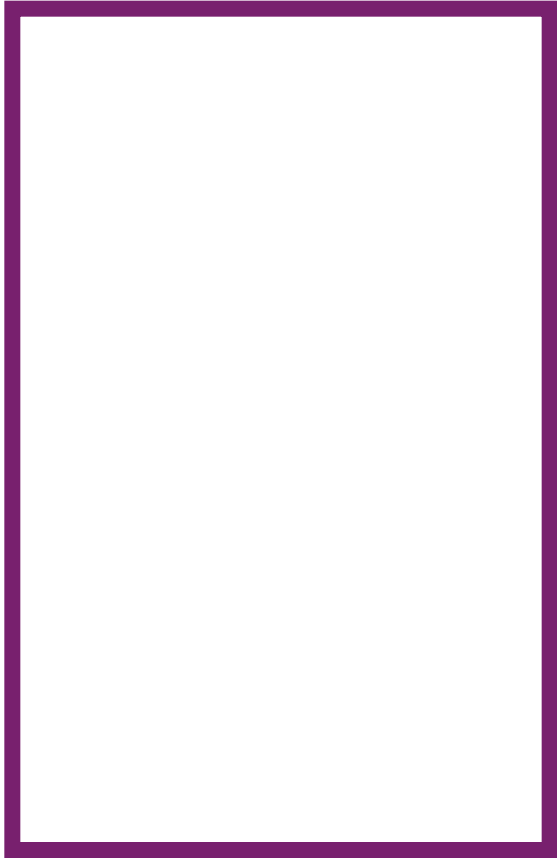
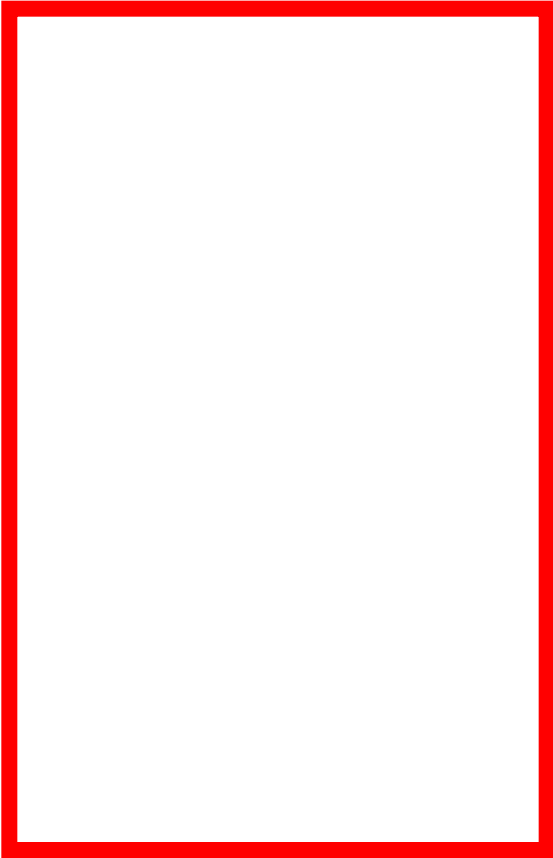
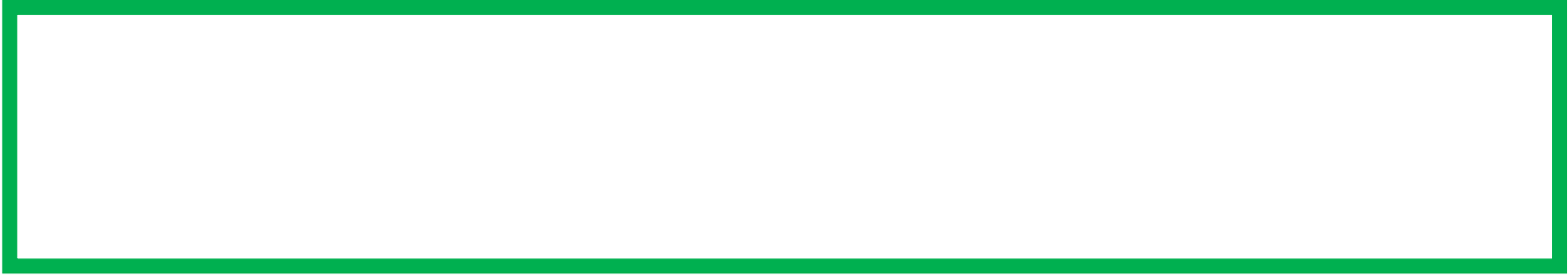
- What happens if Firm 2 does not play the strategy we just found – entering or not with 50% probability?
- Let's say that Firm 2 wants to enter with a 75% probability – **how does this affect Firm 1's incentives?**

Why is this a Nash equilibrium?

- What happens if Firm 2 does not play the strategy we just found – entering or not with 50% probability?
- Let's say that Firm 2 wants to enter with a 75% probability – **how does this affect Firm 1's incentives?**
- **$E(\text{Firm 1 profit} \mid \text{Firm 1 enters}) = 75\% \times -1 + 25\% \times 1 = -0.50$**
- **$E(\text{Firm 1 profit} \mid \text{Firm does not enter}) = 75\% \times 0 + 25\% \times 0 = 0$**
- Firm 1 would always just prefer not to enter!
- So it would **deviate from its mixed strategy**

	Firm 1 enters	Firm 1 does not enter
	Probability that firm 1 enters = p_1	Probability that firm 1 does not enter = $(1 - p_1)$
Firm 2 enters Probability that firm 2 enters = p_2	F1 profit = -1 F2 profit = -1	F1 profit = 0 F2 profit = 1
Firm 2 does not enter Probability that firm 2 does not enter = $(1 - p_2)$	F1 profit = 1 F2 profit = 0	F1 profit = 0 F2 profit = 0





	Pitcher goes high	Pitcher goes low
Batter goes high	Pitcher: 0 Batter: 1	Pitcher: 1 Batter: 0
Batter goes low	Pitcher: 1 Batter: 0	Pitcher: 0 Batter: 2